Prediction of Water Quality Index in Great Lakes

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***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

The Great Lakes are an important part of the water supply and environment of the USA and Canada. As such, being able to accurately predict their water quality is important in ensuring the health of the Great Lakes. To this end, we will design a variety of different machine learning models capable of predicting the water quality index of the Great Lakes based on water quality data from the Great Lakes.

With these models, sudden changes in the Great Lakes' water quality can be caught quickly, allowing action to be taken to correct the change just as quickly. With this rapid response time, the water quality, which the ecosystem in the Great Lakes relies on, can be more efficiently preserved.

II. Related Work

Miller et. al. used both statistical and ML approaches to predict the water quality, using data collected over a span of 20 years. They implemented Multiple Linear Regression (MLR), Artificial Neural Network (ANN), and a Random Forest (RF) algorithm to predict the water quality. Their study showed that the RF algorithm had the best performance, and also showed the importance of Biological Oxygen Demand (BOD) and Turbidity to predict water quality [1].

Hussein et. al. used the classification learner tool in MATLAB to choose the best decision tree model among support vector machines (SVM), k-nearest neighbors (K-NN), ensemble tree (ET), and discrimination analysis (DA). Their results indicate that linear SVM performed the best, and can predict the water quality index with good accuracy [2].

The study by Trebitz et al.(2007) analyzed the water quality of various 58 coastal areas in the Great lakes basin where they analyzed the agriculture disturbance, nutrient concentrations, and suspended particulates and found that Lake Erie wetlands showed greatest nutrients but lowest clarity. They used the Great Lakes Ecological Indicators consortium(GLEI) method. A PCA based index summarized patterns perfectly but other type classification failed to explain variation [3].

While Miller et. al. did create a means to predict water quality, they did so on a reservoir in Poland [1]. We build on this by doing the same for the Great Lakes, which are much larger bodies of water. Similarly, Hussen et. al. performed water quality prediction on groundwater [2], so we will perform a similar process using the Great Lakes instead. Trebitz et. al. did use Great Lakes data, but calculated the water quality using the samples they collected instead of using an ML approach [3]. We will be using an ML approach to predicting water quality, but without the data Trebitz had access to, we use the "Great Lakes Water Quality Monitoring and Surveillancce Data" provided by Datastream instead [5].

III. Methodology

A. Data pre-processing

The dataset we have used in this study is collected by Environment and Climate Change Canada (ECCC) [5], and was last updated on 2025-09-22. The dataset contains numerous measurements of various factors affecting the water quality in the Great Lakes region. The data currently collected spans 24 years, starting from 2000 up to 2024 in the latest version.

We used a standard data pre-processing pipeline to ensure that the dataset is fit for training our models. The summary is as follows:

- Dropped fully null columns, and columns with more than half missing
- Applied KNN imputation
- Calculate WQI indices and created the target column
- Skewness correction with Box-Cox
- Outlier detection and winsorization

The original dataset contained all the predictors in the *CharacteristicName* column, and the values in the *ResultValue*

column. We pivoted the dataset such that the columns are the predictor variables.

The main problem with this dataset is that it is quite sparse. We dropped all the columns that were completely empty, along with those columns where more than half of the rows were missing. After this, we scaled the predictors using the *StandardScaler()* and applied KNN imputation to fill in those missing values. The number of neighbors parameter was chosen as 10.

After this, we calculated the WQI for each row using the weighted formula,

$$WQI = \frac{\sum_{i=1}^{n} w_i \cdot q_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

where w_i is the weight of the i-th parameter, q_i is the quality rating of the i-th parameter, and n is the number of parameters.

We checked the skewness and found out that the dataset had a right-skewed distribution. We used the Box-Cox transformation to correct the skewed distribution.

Additionally, we detected outliers using the Z-score. Roughly 7% of the dataset had a Z-score greater than 3 or less than -3, and were classified as outliers. Since we had a small dataset, we opted for winsorization and capped the ranges of those outliers.

We also checked the correlation of the predictors. However, we decided not to use any filtering even though some predictors were highly correlated.

After all the preprocessing steps, we had the following dataset:

- 6550 observations
- 17 predictor columns
- 1 target variable (WQI)
- 3 index columns

Since the dataset only contains about 6500 observations, we decided to use a smaller testing and validation split. We created a 90/5/5 split for training, testing and validation sets respectively.

B. Linear models implemented

1) Linear Regression: We started with the simple Multiple Linear Regression (MLR) model using the *sklearn* library.

We used 5-fold cross-validation to fit the model and extracted the cross-validation results.

2) Support Vector Regression: Support Vector Regression (SVR) model was implemented using the Radial-Basis Function (RBF) kernel.

We used 5-fold cross-validation to fit the model across a grid of (C, $epsilon$, $gamma$) and extracted the best model.

3) Neural Network: A Neural Network (NN) model was constructed with the following parameters:

- 1 input layer
- 3 hidden layers, ReLU activation
- 1 output layer, linear activation
- L2 regularization
- Xavier Normal initialization
- Adam optimizer, learning rate = $1 \cdot 10^{-3}$
- Mean Squared Error loss function
- 100 epochs

C. Model evaluation

In order to evaluate the models, we used the standard metrics - Coefficient of determination (R^2), Root-Mean Squared Error ($RMSE$) and Mean Absolute Error (MAE).

We calculated these metrics using the results from the test set, and then plotted the predicted WQI against true WQI values.

For the neural network, we also plotted the Loss and Accuracy curves to evaluate the training process.

IV. Experimental Results

The cross-validation results reported in Table I suggests that the SVR model had the better performance when compared to the MLR model.

When taking a look at the test-set results reported in Table II, we can see that the neural network model had the best performance among all the models tested. If we consider MAE as the testing metric, then the SVR model comes on top.

TABLE I
Cross-Validation Results of MLR and SVR Models

Model	R^2	RMSE	MAE
MLR	0.7864	3.0003	1.7542
SVR	0.9646	1.2383	0.2645

TABLE II
Testing Results of MLR, SVR and NN Models

Model	R^2	RMSE	MAE
MLR	0.8165	2.4582	1.5674
SVR	0.9595	1.1550	0.2738
NN	0.9808	0.7950	0.4968

We also used the *permutation_importance* function in *sklearn* library to determine the most important predictors in each model. The results are listed in the Tables III through V. The actual importance values for the predictors are not reported here.

The most important predictors across all the models can be extracted as follows (not ranked by importance):

- Dissolved Oxygen (DO)
- Temperature, water
- Chloride
- Fluoride
- Chlorophyll a, uncorrected for pheophytin

TABLE III
Top 3 Important Predictors for MLR model

Rank	MLR
1	Dissolved oxygen (DO)
2	Temperature, water
3	Chlorophyll a, uncorrected for pheophytin

TABLE IV
TOP 3 IMPORTANT PREDICTORS FOR SVR MODEL

Rank	SVR
1	Chloride
2	Fluoride
3	Dissolved oxygen (DO)

TABLE V
TOP 3 IMPORTANT PREDICTORS FOR NN MODEL

Rank	NN
1	Chloride
2	Dissolved oxygen (DO)
3	Temperature, water

We plotted the model predictions and then compared it with the perfect (linear) prediction for each of the models. The results are shown in Figure 1 (MLR), Figure 3 (SVR) and Figure 4 (NN). The residuals for the MLR model is also plotted in Figure 2. Similarly, we plotted the loss and error curves for the NN model in Figures 5 and 6.

From the results, we can see that all three models have good performance and can reasonably approximate the linear relationship of the target variable (WQI). The MLR model has more errors compared to the other two. While looking at the graphs for SVR and NN, we can see that SVR has lower error (which explains the lower MAE), but it has some bad predictions that are lowering the other metrics.

If we take a look at the MLR residual plot, we can observe that the model predictions mostly range between -5 to 5 of the actual predictions (except the two outliers).

We also plotted the graphs with the training errors and validation errors. Since the error was high at the start of the training, we opted to use a log scale for the Y-axis. Both graphs show that the neural network trained well and performed optimally on the validation set. The spikes on the validation curves are most likely caused by the small validation set.

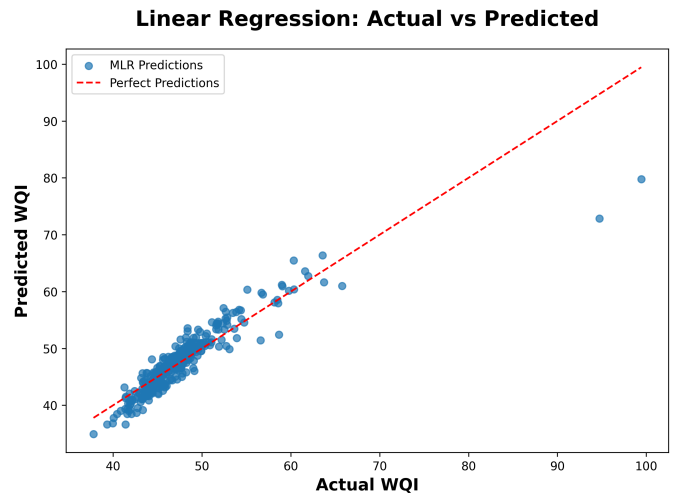

Fig. 1. Linear regression actual vs predicted WQI.

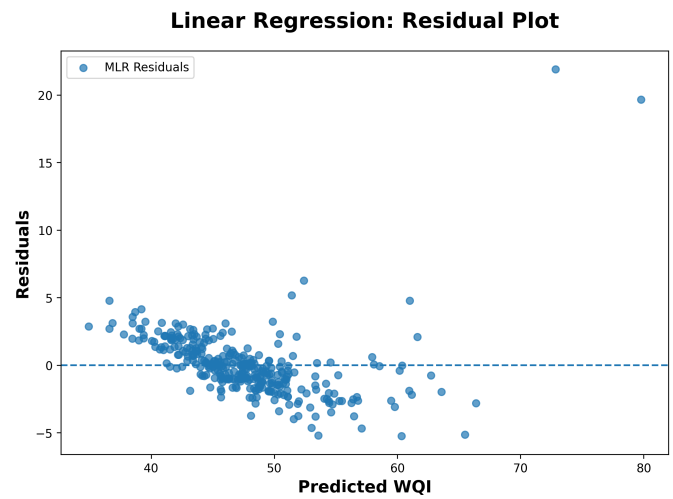

Fig. 2. Linear regression residuals.

V. DISCUSSION AND CONCLUSION

This study implemented the three models, Multiple Regression Model (MLR), Support Vector Regression (SVR), and Neural Networks (NN) and the performance of those three models illustrate the capability of machine learning approaches to analyze the relationship between water quality parameters and the Water Quality Index (WQI) in the Great Lakes basin datasets.

The comparison between the three models reveals a clear difference in their prediction capability. The MLR model performed an R^2 of 0.8165, RMSE of 2.4582, and MAE of 1.5674 on the test set, which indicate the moderate predictive skill. On the other hand, the SVR model performed significantly well as compared to MLR with an R^2 of 0.9595 and RMSE of 1.1550 with MAE of 0.2738, which is also the lowest MAE. The Neural Network achieved the best accuracy overall with R^2 of 0.9808 and the lowest of RMSE of 0.7950, but its MAE(0.4968) was higher than that of SVR.

The quantitative result in Figure 1 are consistent with the qualitative pattern observed in the prediction plot. The MLR predictions highlight significant depression around the 1:1 reference line, highlighting systematic deviations from actual WQI values. This is further strongly supported by the residual distribution plot (Fig. 2), where the majority of errors lie within the range of ±5 units. This illustrates that while MLR captures the general trend, it somehow lacks some flexibility to model nonlinear relationships in the data.

The SVR model (Fig. 3) demonstrates a tighter clustering of predictions along the ideal line, indicating its strong performance in minimizing average error. However, some random points deviate significantly from the trend. Although the numbers are small, this explains that despite its low MAE, its RMSE and R^2 are slightly lower than the Neural Network. This reflects that SVR is efficient in capturing the structure of the data but is more sensitive to local prediction errors.

The Neural Network results (Fig. 4) demonstrate the closest

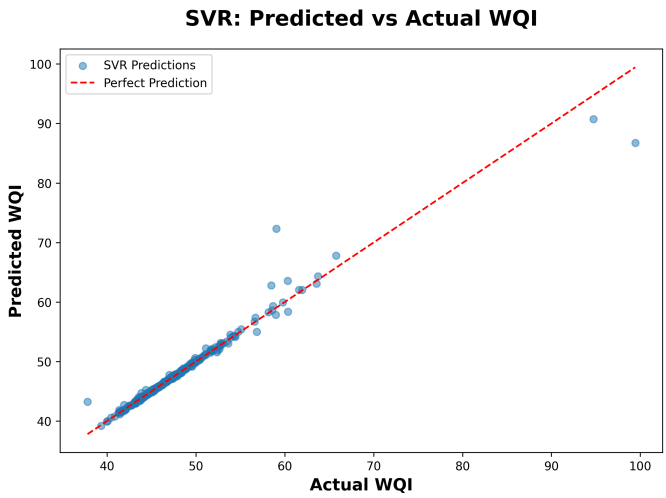

Fig. 3. Support vector regression actual vs predicted WQI.

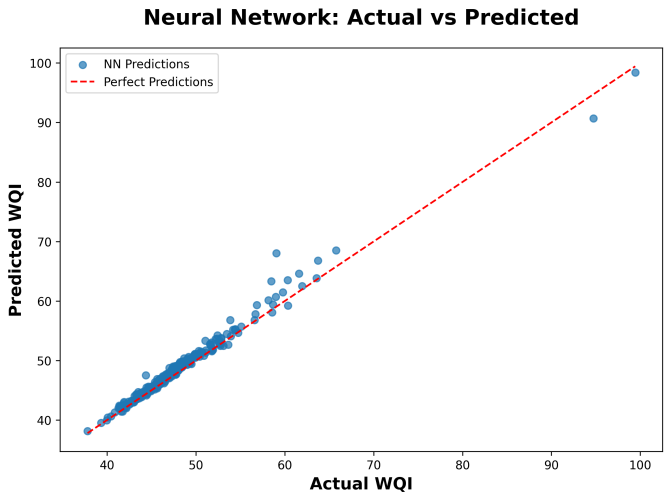

Fig. 4. Neural network actual vs predicted WQI.

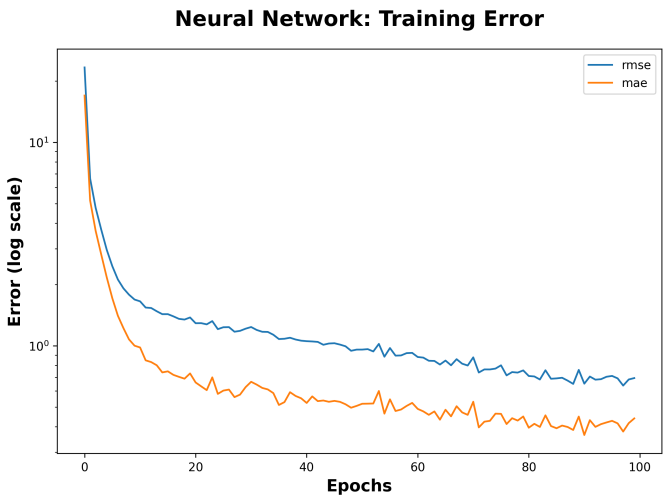

Fig. 5. Neural network training errors.

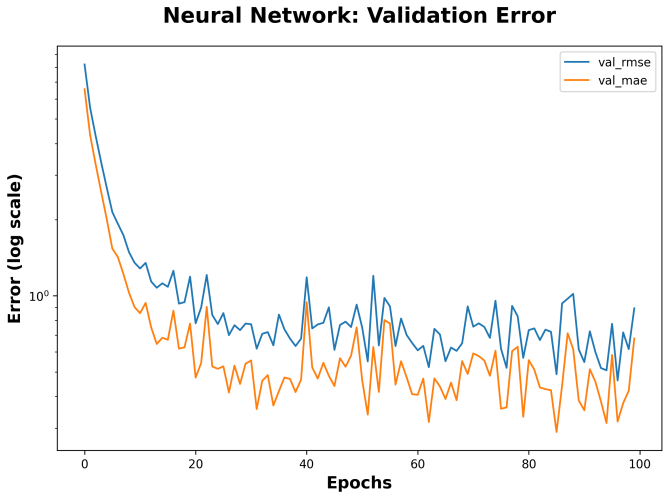

Fig. 6. Neural network validation errors.

alignment between the predicted and actual WQI values, with the closest dispersion across the full range. Its superior R^2 and RMSE performance highlights the consistency of this model. The training and validation curves (Fig. 5 and Fig. 6) demonstrate an accurate and consistent decrease in error over epochs, indicating stable convergence. The fluctuations in the validation curve are consistent with the small validation datasets, rather than instability in the model.

From the data perspective, it is clear that the preprocessing pipeline significantly influenced the performance of the model. The application of KNN imputation solved the sparsity of the datasets, while the Box-Cox transformation decreased skewness and improved distributional properties. The consistent patterns across the models highlight the importance of predictors analysis. Some variables were identified as key contributors, such as dissolved oxygen, water temperature, chloride, fluoride, and chlorophyll-a. The result across MLR, SVR, and NN indicates that these parameters are the key variables in determining WQI within the datasets and support the modeling approach. For this problem, the overall results highlight that the nonlinear models provide substantial improvements over linear methods for error reduction and the ability to showcase the complex relationships between the datasets.

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